KAVACH · GeoSatya Smart Dispatch & Civic Resolution: Handoff

Repository **https://github.com/ananyadhagat/Kavach-epam** · branch **geosatya-location-verification** · base commit **9053d0c** · module `services/geosatya` · 2026-10-30

> **State.** All 14 checkpoints are complete and verified: **156 tests pass**, the live localhost run passed, and the browser checks passed. This phase's changes are **not committed or pushed**; the user handles git. The Dispatch agent's ranking file and GeoSatya's geofence rule were not modified.

A. Team summary

GeoSatya now runs a civic incident from report to resolution like a delivery-tracking app, adapted to civic work:

- A report is saved, analysed by Incident Intelligence (priority) and dispatched with the Dispatch agent's rules.
- The worker accepts or declines; a decline sends it to the next eligible worker.
- The worker starts the journey and shares their location. The citizen sees the phase, live or stale position, an approximate ETA range and the ETR.
- Arrival is confirmed from two good readings. The work is still verified by the server's 15 m geofence against the original location, then by a supervisor.
- Supervisors get a dispatch board flagging unassigned, unaccepted, stale, stationary, overdue and awaiting-review incidents, and can cancel, re-dispatch, reassign or assign.
- Every timeline entry comes from recorded events.

B. Technical summary

- **Architecture:** GeoSatya (FastAPI + SQLite, vanilla JS) is the system of record. The Incident Intelligence agent is a separate HTTP service (:8001). The Dispatch agent's ranking function is used in-process with GeoSatya's workers.
- **Dispatch:** eligibility (availability, capacity, skills, equipment, location ≤ 15 min old, within 2 km), then 0.7·distance + 0.3·workload. Released workers are excluded. There is a labelled workload fallback or escalation, and one task per incident (no duplicates).
- **Acceptance:** `task_progress.accepted_at`; starting the journey or work implies it. A timeout is flagged after 15 min (no background scheduler). Release kinds are DECLINED / ACCEPTANCE_TIMEOUT / CANCELLED, allowed only before any completion attempt.
- **Tracking:** only the assigned worker, only for an active task. Points are validated, throttled to 10 s, capped at 20 and deleted when field work ends. Viewers poll (15 s incident page, 30 s board). The citizen sees the position only while it is live.
- **ETA:** straight line × 1.4 at the observed approach speed (or an assumed 20 km/h), shown as a minute range. Stale and stationary states are flagged. No route is drawn because there is no routing service.
- **Arrival:** 2 consecutive readings within 50 m, accuracy ≤ 50 m, fix ≤ 120 s old. This is separate from the geofence.
- **ETR:** recomputed on events: travel allowance or ETA, plus work (median of resolved jobs or a category baseline), plus a review buffer. Actual durations come from timestamps.
- **Verification:** unchanged server Haversine against the task's stored target: PASS / FAIL / REVIEW_REQUIRED / NOT_CHECKED, followed by supervisor review.
- **Database:** the new nullable column is added by an additive `ALTER TABLE` at start-up. It was verified on the existing local database with all incidents kept.

Status mapping (no new status enums)

Conceptual state	Representation
ASSIGNED	task ASSIGNED, no accepted_at
ACCEPTED	task ASSIGNED + task_progress.accepted_at
EN_ROUTE	task ASSIGNED + tracking_started_at
ARRIVED	task_progress.arrived_at (confirmed readings)
IN_PROGRESS	task IN_PROGRESS
COMPLETION_SUBMITTED / VERIFICATION / REVIEW	task AWAITING_VERIFICATION
RESOLVED	task RESOLVED
REJECTED_OR_MORE_EVIDENCE	back to ASSIGNED, attempt + 1
DECLINED / TIMEOUT / CANCELLED	task removed, dispatch record RELEASED, incident SUBMITTED, re-dispatch

C. Checkpoint report

#	Checkpoint	Status	Note
1	Inspect repository, ZIP and branches	completed	
2	Compare GeoSatya, Incident Intelligence and Dispatch Work	completed	
3	Connect reporting to Incident Intelligence	completed	previous commit
4	Connect incidents to dispatch and assignment	completed	previous commit
5	Worker selection	completed	previous commit; released-worker exclusion added
6	Acceptance and journey status	completed	
7	Authorised live tracking	completed	
8	ETA and arrival detection	completed	
9	Work start and ETR	completed	previous commit
10	Completion evidence and geofence verification	completed	unchanged rule
11	Citizen timeline and supervisor dashboard	completed	
12	Tests and integration fixes	completed	
13	Localhost verification	completed	
14	Documentation and handoff	completed	commit and push left to the user

Integration status

Area	Status	Phase	Details
incident intelligence	completed	previous commit 9053d0c	HTTP call after the report is committed; stable X-Request-ID; output stored separately; severity -> priority; failures never lose the report
dispatch and worker selection	completed	previous commit 9053d0c	Dispatch agent ranking reused unchanged in this phase (eligibility + 70% distance / 30% workload); Labelled workload fallback or escalation; Released workers excluded from automatic re-dispatch (new)

acceptance and journey	completed	this phase	POST /tasks/{id}/accept, /decline; starting the journey or work implies acceptance; Acceptance timeout flagged after ACCEPTANCE_TIMEOUT_MIN; supervisor re-dispatch via POST /tasks/{id}/release kind ACCEPTANCE_TIMEOUT; Supervisor cancellation via POST /tasks/{id}/release kind CANCELLED; No duplicate active assignments: one task per incident, released task removed before re-dispatch; No background scheduler: timeouts are flagged, not auto-reassigned
live worker tracking	completed	previous commit + this phase	Authorised assigned worker only, active task only; citizen sees position only while live; Points capped and deleted when field work ends, on reassignment or release; Controlled polling: incident page 15 s, board 30 s, visible tab only
eta	completed	previous commit + this phase	Approximate straight-line x 1.4 at observed or assumed speed, shown as a range; States WAITING_FOR_LOCATION, AVAILABLE, STALE, ARRIVED, NOT_APPLICABLE; Stationary worker flag (new); No road routing and no drawn route
arrival detection	completed	this phase	2 consecutive readings within 50 m, accuracy <= 50 m, fix not older than 120 s; Separate from the 15 m completion geofence
etr and actual resolution	completed	previous commit	Event-based ETR with category baseline or historical median; Actual durations from recorded timestamps
completion verification	completed	unchanged GeoSatya	Server Haversine against the task's stored target; PASS / FAIL / REVIEW_REQUIRED / NOT_CHECKED unchanged
citizen timeline	completed	this phase	progress.events from persisted rows only, including withdrawn and pending assignments; Phase text: UNASSIGNED .. RESOLVED
supervisor dispatch board	completed	this phase	GET /dispatch/board and the Dispatch Board page with attention flags and filters
database	completed	this phase	New nullable column task_progress.accepted_at; init_schema adds missing nullable columns (additive ALTER TABLE); verified on the existing local database with all 4 incidents kept
ratings	not_started		Not in any branch; proposal in docs/INTEGRATION.md section 8

D. Testing report

Executed:

- pytest services/geosatya full suite
- Live HTTP run against GeoSatya :8000 and the Incident Intelligence agent :8001 (mock mode)
- Browser: worker accept, supervisor cancel assignment, dispatch board, citizen timeline; desktop and 375 px

Passed:

- 156 pytest tests passed (110 original + 29 integration + 17 scenario), 1 dependency deprecation warning
- Scenario A successful resolution (automated + live run)
- Scenario B no suitable worker
- Scenario C decline re-dispatch without duplicates; acceptance timeout flag and re-dispatch (automated + live decline)
- Scenario D location unavailable
- Scenario E stale location and resume
- Scenario F arrival needs confirmed fresh readings (automated + live)
- Scenario G completion outside geofence is FAIL and not accepted
- Scenario H review pending is not shown as resolved (automated + live)
- Scenario I ETR and tracking unavailable when unassigned

- Scenario J cancellation stops tracking and new worker starts clean (automated + browser)
- Column migration on an older database (automated + real local database)
- Browser: no console errors after the map-animation fix; no horizontal overflow at 375 px

Failed: none. **Skipped:** none.

Blocked / not run:

- Real phone GPS and camera (browser pane blocks them; labelled simulation used)
- Gemini mode of the intelligence agent (no API key)
- Partner branches' own test suites (not run)

Bugs found during verification and fixed:

- Legacy tasks tracked before accepted_at existed were flagged as not accepted in time
- Leaflet error when a page re-rendered while the map was animating a zoom
- Audit view labelled dispatch and intelligence actions as 'Verification engine'

Scenarios A–J

	Scenario	Evidence	Result
A	Successful resolution	`test_scenario_a_successful_resolution + live run`	Passed
B	No suitable worker	`test_scenario_b_no_suitable_worker`	Passed
C	Worker declines / misses assignment	`test_scenario_c_decline_redispatches_without_duplicates, test_acceptance_timeout_is_flagged_and_can_be_redispatched + live decline`	Passed
D	Worker location unavailable	`test_scenario_d_location_unavailable`	Passed
E	Stale location	`test_scenario_e_stale_location_is_labelled_and_flagged`	Passed
F	Worker reaches the site	`test_scenario_f_arrival_needs_confirmed_fresh_readings + live run`	Passed
G	Completion outside the geofence	`test_scenario_g_and_h_outside_geofence_then_review_pending`	Passed
H	Work done, review pending	`test_scenario_g_and_h_outside_geofence_then_review_pending + live run`	Passed
I	ETA / ETR cannot be calculated	`test_scenario_b_no_suitable_worker, test_scenario_d_location_unavailable`	Passed
J	Assignment cancelled / reassigned	`test_scenario_j_cancel_stops_tracking_and_new_worker_starts_clean, test_reassignment_stops_tracking_and_moves_access + browser`	Passed

Live localhost run (captured output)

```
A1 report GS-TQ7HB5 | intelligence COMPLETED HIGH priority HIGH
A2 dispatch DISPATCH_RANKING -> W-DEMO-15 | phase AWAITING_ACCEPTANCE
C decline by W-DEMO-15 200 -> re-dispatched to W-DEMO-12 | phase AWAITING_ACCEPTANCE
accept 200
ping 400 m {'stored': True, 'distance_m': 399.6, 'arrived': False} | citizen: EN_ROUTE LIVE AVAILABLE 1 3
ping 30 m {'stored': True, 'distance_m': 30.0, 'arrived': False} | citizen: EN_ROUTE LIVE AVAILABLE 1 2
ping 20 m {'stored': True, 'distance_m': 20.0, 'arrived': True} | citizen: ARRIVED LIVE ARRIVED None None
H completion PASS 5.0 m | citizen status AWAITING_VERIFICATION AWAITING_REVIEW
board AWAITING_REVIEW ['AWAITING_REVIEW'] | counts {'UNASSIGNED': 0, 'ACCEPTANCE_OVERDUE': 2,
```

```
'TRACKING_STALE': 1, 'WORKER_STATIONARY': 0, 'ETR_OVERDUE': 1, 'AWAITING_EVIDENCE': 0, 'AWAITING_REVIEW': 2}
final RESOLVED RESOLVED {'total_min': 0.4, 'to_assignment_min': 0.0, 'travel_min': 0.4, 'work_min': 0.0,
'review_min': 0.0, 'reported_at': '2026-10-29T19:37:20.966542Z', 'resolved_at': '2026-10-29T19:37:46.206032Z'}
timeline ['REPORTED', 'ANALYSED', 'ASSIGNED', 'ASSIGNMENT_RELEASED', 'ASSIGNED', 'ACCEPTED', 'EN_ROUTE',
'ARRIVED', 'WORK_STARTED', 'EVIDENCE_SUBMITTED', 'LOCATION_VERIFIED', 'REVIEWED', 'RESOLVED']
```

E. Localhost instructions

- Optional agent: in a checkout of incident-intelligence-agent, $env:KAVACH_LLM_MODE='mock'; python -m uvicorn kavach.main:app --app-dir src --port 8001
- GeoSatya: cd services/geosatya; optionally $env:INTELLIGENCE_URL='http://127.0.0.1:8001'; uvicorn geosatya.main:create_app --factory --reload
- Open http://localhost:8000; pick a demo role; enable Simulate GPS (demo) when there is no GPS

URLs: `http://localhost:8000`, `http://localhost:8000/docs`, `http://localhost:8000/health`, `http://127.0.0.1:8001/health`. Location and camera work on `http://localhost`; phones need HTTPS. Simulated GPS is always labelled "Simulated".

F. Known limitations

- No road routing; ETA is approximate
- Browser cannot track in the background
- ETR baselines are placeholders until history exists
- Dispatch has no service-area or location-accuracy rule (no such data)
- Released assignments delete the task row (history kept in dispatch_records and audit); not allowed after evidence exists
- incident-intelligence-agent requirements.txt lacks python-dotenv; dispatch-work requirements.txt is empty and app/_init_.py is misnamed
- Demo identities are not authentication

G. Continuation guide

1. `cd D:\Desktop\Kavach-epam-team`, then `git status`. Expect only `services/geosatya/` changes.
2. `cd services\geosatya`, then `python -m pytest`. Expect 156 passed.
3. Run the app as in section E, and follow the README "Integrated demo".
4. Commit with explicit paths, then push to `origin geosatya-location-verification`. Never push to main.
5. Pending: Commit and push (user handles git); Ratings (team decision); Real-phone HTTPS test; Optional routing provider; Admin screen for worker skills; Background job for acceptance timeouts if wanted.

H. JSON report


```
{
  "project": "Kavach GeoSatya: smart dispatch, live worker tracking, ETA and civic incident resolution",
  "generated": "2026-10-30",
  "repository": {
    "url": "https://github.com/ananyadhagat/Kavach-epam",
    "branch": "geosatya-location-verification",
    "base_commit": "9053d0c",
    "module_path": "services/geosatya"
  },
  "sources_inspected": {
    "geosatya": "origin/geosatya-location-verification 9053d0c (previous integration already committed by the user)",
    "incident_intelligence_agent": "origin/incident-intelligence-agent e2ff0e9, unchanged since the previous inspection; ZIP
identical",
    "dispatch_work": "origin/dispatch-work 6ef3bf9, unchanged; ZIP identical",
```

```
    "zip_note": "No new ZIP was attached to this request; the two ZIPs in D:/Downloads are the same files inspected before"
  },
  "integration": {
    "incident_intelligence": {
      "status": "completed",
      "phase": "previous commit 9053d0c",
      "details": [
        "HTTP call after the report is committed; stable X-Request-ID; output stored separately; severity -> priority; failures
never lose the report"
      ]
    },
    "dispatch_and_worker_selection": {
      "status": "completed",
      "phase": "previous commit 9053d0c",
      "details": [
        "Dispatch agent ranking reused unchanged in this phase (eligibility + 70% distance / 30% workload)",
        "Labelled workload fallback or escalation",
        "Released workers excluded from automatic re-dispatch (new)"
      ]
    },
    "acceptance_and_journey": {
      "status": "completed",
      "phase": "this phase",
      "details": [
        "POST /tasks/{id}/accept, /decline; starting the journey or work implies acceptance",
        "Acceptance timeout flagged after ACCEPTANCE_TIMEOUT_MIN; supervisor re-dispatch via POST /tasks/{id}/release kind
ACCEPTANCE_TIMEOUT",
        "Supervisor cancellation via POST /tasks/{id}/release kind CANCELLED",
        "No duplicate active assignments: one task per incident, released task removed before re-dispatch",
        "No background scheduler: timeouts are flagged, not auto-reassigned"
      ]
    },
    "live_worker_tracking": {
      "status": "completed",
      "phase": "previous commit + this phase",
      "details": [
        "Authorised assigned worker only, active task only; citizen sees position only while live",
        "Points capped and deleted when field work ends, on reassignment or release",
        "Controlled polling: incident page 15 s, board 30 s, visible tab only"
      ]
    },
    "eta": {
      "status": "completed",
      "phase": "previous commit + this phase",
      "details": [
        "Approximate straight-line x 1.4 at observed or assumed speed, shown as a range",
        "States WAITING_FOR_LOCATION, AVAILABLE, STALE, ARRIVED, NOT_APPLICABLE",
        "Stationary worker flag (new)",
        "No road routing and no drawn route"
      ]
    },
    "arrival_detection": {
      "status": "completed",
      "phase": "this phase",
      "details": [
        "2 consecutive readings within 50 m, accuracy <= 50 m, fix not older than 120 s",
        "Separate from the 15 m completion geofence"
      ]
    },
    "etr_and_actual_resolution": {
      "status": "completed",
      "phase": "previous commit",
      "details": [
        "Event-based ETR with category baseline or historical median",
        "Actual durations from recorded timestamps"
      ]
    },
    "completion_verification": {
      "status": "completed",
      "phase": "unchanged GeoSatya",
```

```
      "details": [
        "Server Haversine against the task's stored target; PASS / FAIL / REVIEW_REQUIRED / NOT_CHECKED unchanged"
      ]
    },
    "citizen_timeline": {
      "status": "completed",
      "phase": "this phase",
      "details": [
        "progress.events from persisted rows only, including withdrawn and pending assignments",
        "Phase text: UNASSIGNED .. RESOLVED"
      ]
    },
    "supervisor_dispatch_board": {
      "status": "completed",
      "phase": "this phase",
      "details": [
        "GET /dispatch/board and the Dispatch Board page with attention flags and filters"
      ]
    },
    "database": {
      "status": "completed",
      "phase": "this phase",
      "details": [
        "New nullable column task_progress.accepted_at",
        "init_schema adds missing nullable columns (additive ALTER TABLE); verified on the existing local database with all 4
incidents kept"
      ]
    },
    "ratings": {
      "status": "not_started",
      "details": [
        "Not in any branch; proposal in docs/INTEGRATION.md section 8"
      ]
    }
  },
  "status_mapping": {
    "ASSIGNED": "task ASSIGNED, no accepted_at",
    "ACCEPTED": "task ASSIGNED + task_progress.accepted_at",
    "EN_ROUTE": "task ASSIGNED + tracking_started_at",
    "ARRIVED": "task_progress.arrived_at (confirmed readings)",
    "IN_PROGRESS": "task IN_PROGRESS",
    "COMPLETION_SUBMITTED / VERIFICATION / REVIEW": "task AWAITING_VERIFICATION",
    "RESOLVED": "task RESOLVED",
    "REJECTED_OR_MORE_EVIDENCE": "back to ASSIGNED, attempt + 1",
    "DECLINED / TIMEOUT / CANCELLED": "task removed, dispatch record RELEASED, incident SUBMITTED, re-dispatch"
  },
  "testing": {
    "tests_run": [
      "pytest services/geosatya full suite",
      "Live HTTP run against GeoSatya :8000 and the Incident Intelligence agent :8001 (mock mode)",
      "Browser: worker accept, supervisor cancel assignment, dispatch board, citizen timeline; desktop and 375 px"
    ],
    "passed": [
      "156 pytest tests passed (110 original + 29 integration + 17 scenario), 1 dependency deprecation warning",
      "Scenario A successful resolution (automated + live run)",
      "Scenario B no suitable worker",
      "Scenario C decline re-dispatch without duplicates; acceptance timeout flag and re-dispatch (automated + live decline)",
      "Scenario D location unavailable",
      "Scenario E stale location and resume",
      "Scenario F arrival needs confirmed fresh readings (automated + live)",
      "Scenario G completion outside geofence is FAIL and not accepted",
      "Scenario H review pending is not shown as resolved (automated + live)",
      "Scenario I ETR and tracking unavailable when unassigned",
      "Scenario J cancellation stops tracking and new worker starts clean (automated + browser)",
      "Column migration on an older database (automated + real local database)",
      "Browser: no console errors after the map-animation fix; no horizontal overflow at 375 px"
    ],
    "failed": [],
    "skipped": [],
    "blocked": [
```

```
      "Real phone GPS and camera (browser pane blocks them; labelled simulation used)",
      "Gemini mode of the intelligence agent (no API key)",
      "Partner branches' own test suites (not run)"
    ],
    "bugs_found_and_fixed": [
      "Legacy tasks tracked before accepted_at existed were flagged as not accepted in time",
      "Leaflet error when a page re-rendered while the map was animating a zoom",
      "Audit view labelled dispatch and intelligence actions as 'Verification engine'"
    ]
  },
  "localhost": {
    "status": "verified",
    "startup_instructions": [
      "Optional agent: in a checkout of incident-intelligence-agent, $env:KAVACH_LLM_MODE='mock'; python -m uvicorn kavach.main:app
--app-dir src --port 8001",
      "GeoSatya: cd services/geosatya; optionally $env:INTELLIGENCE_URL='http://127.0.0.1:8001'; uvicorn geosatya.main:create_app --
factory --reload",
      "Open http://localhost:8000; pick a demo role; enable Simulate GPS (demo) when there is no GPS"
    ],
    "urls": [
      "http://localhost:8000",
      "http://localhost:8000/docs",
      "http://localhost:8000/health",
      "http://127.0.0.1:8001/health"
    ],
    "ports": [
      8000,
      8001
    ]
  },
  "git": {
    "files_changed": [
      "services/geosatya/.env.example",
      "services/geosatya/README.md",
      "services/geosatya/docs/INTEGRATION.md",
      "services/geosatya/docs/GEOSATYA_INTEGRATION_HANDOFF.md",
      "services/geosatya/frontend/js/app.js",
      "services/geosatya/frontend/js/live.js",
      "services/geosatya/frontend/js/maps.js",
      "services/geosatya/frontend/js/pages/tasks.js",
      "services/geosatya/frontend/styles.css",
      "services/geosatya/geosatya/api/routes.py",
      "services/geosatya/geosatya/api/schemas.py",
      "services/geosatya/geosatya/api/views.py",
      "services/geosatya/geosatya/config.py",
      "services/geosatya/geosatya/db.py",
      "services/geosatya/geosatya/models.py",
      "services/geosatya/geosatya/services/common.py",
      "services/geosatya/geosatya/services/dispatch.py",
      "services/geosatya/geosatya/services/estimates.py",
      "services/geosatya/geosatya/services/tasks.py",
      "services/geosatya/geosatya/services/tracking.py",
      "services/geosatya/tests/test_integration.py"
    ],
    "files_added": [
      "services/geosatya/frontend/js/pages/board.js",
      "services/geosatya/geosatya/services/board.py",
      "services/geosatya/tests/test_scenarios.py",
      "services/geosatya/docs/GeoSatya_Smart_Dispatch_Report.json",
      "services/geosatya/docs/GeoSatya_Smart_Dispatch_Handoff.pdf"
    ],
    "untracked_from_earlier_request": [
      "services/geosatya/docs/GeoSatya_Location_Verification_Outputs.pdf",
      "services/geosatya/docs/GeoSatya_Location_Verification_Samples.json"
    ],
    "not_modified": [
      "services/geosatya/geosatya/dispatch/ranking.py (Dispatch agent rules)",
      "services/geosatya/geosatya/location/geofence.py (geofence rule)"
    ],
    "commit_created": false,
```

```
  "push_completed": false,
  "target_branch": "geosatya-location-verification"
},
"checkpoints": [
  {
    "id": 1,
    "name": "Inspect repository, ZIP and branches",
    "status": "completed"
  },
  {
    "id": 2,
    "name": "Compare GeoSatya, Incident Intelligence and Dispatch Work",
    "status": "completed"
  },
  {
    "id": 3,
    "name": "Connect reporting to Incident Intelligence",
    "status": "completed",
    "note": "previous commit"
  },
  {
    "id": 4,
    "name": "Connect incidents to dispatch and assignment",
    "status": "completed",
    "note": "previous commit"
  },
  {
    "id": 5,
    "name": "Worker selection",
    "status": "completed",
    "note": "previous commit; released-worker exclusion added"
  },
  {
    "id": 6,
    "name": "Acceptance and journey status",
    "status": "completed"
  },
  {
    "id": 7,
    "name": "Authorised live tracking",
    "status": "completed"
  },
  {
    "id": 8,
    "name": "ETA and arrival detection",
    "status": "completed"
  },
  {
    "id": 9,
    "name": "Work start and ETR",
    "status": "completed",
    "note": "previous commit"
  },
  {
    "id": 10,
    "name": "Completion evidence and geofence verification",
    "status": "completed",
    "note": "unchanged rule"
  },
  {
    "id": 11,
    "name": "Citizen timeline and supervisor dashboard",
    "status": "completed"
  },
  {
    "id": 12,
    "name": "Tests and integration fixes",
    "status": "completed"
  },
  {
```

```
      "id": 13,
      "name": "Localhost verification",
      "status": "completed"
    },
    {
      "id": 14,
      "name": "Documentation and handoff",
      "status": "completed",
      "note": "commit and push left to the user"
    }
  ],
  "pending": [
    "Commit and push (user handles git)",
    "Ratings (team decision)",
    "Real-phone HTTPS test",
    "Optional routing provider",
    "Admin screen for worker skills",
    "Background job for acceptance timeouts if wanted"
  ],
  "known_issues": [
    "No road routing; ETA is approximate",
    "Browser cannot track in the background",
    "ETR baselines are placeholders until history exists",
    "Dispatch has no service-area or location-accuracy rule (no such data)",
    "Released assignments delete the task row (history kept in dispatch_records and audit); not allowed after evidence exists",
    "incident-intelligence-agent requirements.txt lacks python-dotenv; dispatch-work requirements.txt is empty and app/_init_.py is misnamed",
    "Demo identities are not authentication"
  ],
  "next_steps": [
    "Review the diff and commit with explicit paths under services/geosatya, then push to origin geosatya-location-verification",
    "Team decisions: DISPATCH_FALLBACK default, auto-reassign on acceptance timeout, ratings",
    "Phone test over HTTPS"
  ],
  "handoff_document": "services/geosatya/docs/GEOSATYA_INTEGRATION_HANDOFF.md",
  "pdf_report": "services/geosatya/docs/GeoSatya_Smart_Dispatch_Handoff.pdf"
}
```